Department of
Computer Science and Engineering
Independent University, Bangladesh(IUB)

Experiment No: 01

Experiment Title: Basic characteristics of DIGITAL LOGIC GATES (AND, OR, NOR, NOT, NOR, NAND, XOR) and verification of their truth table.

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Experiment Name	Basic characteristics of DIGITAL LOGIC GATES (AND, OR, NOR, NOT, NOR, NAND, XOR) and verification of their truth table.	
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Course Title	Lab work based on CSE204	
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Section	01	
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2. Objectives:

1. To study the basic digital logic gates: **AND, OR, NOT, NAND, NOR, and XOR**.
2. To understand the operation of these logic gates using **Boolean expressions and truth tables**.
3. To represent the logic functions using **logic diagrams**.
4. To verify the **truth tables** of the different logic gates experimentally.
5. To observe the **pulse response** of the logic gates.

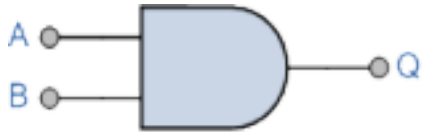
3.Required Components and Equipment:

No.	Component / Equipment	IC Number / Specification	Quantity
1.	AND GATE IC	IC No. 7408	1
2.	OR GATE IC	IC No. 7432	1
3.	NOT GATE IC	IC No. 7404	1
4.	NAND GATE IC	IC No. 7400	1
5.	NOR GATE IC	IC No. 7402	1
6.	XOR GATE IC	IC No. 7486	1
7.	Digital Electronic Trainer		1
8.	Power Supply Unit		1

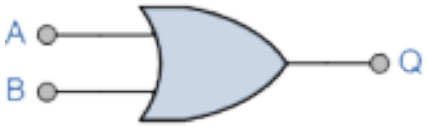
4.Theoretical Background :

- **AND** A multi-input circuit in which the output is 1 only if all inputs are 1. Q=A**.**B
- **OR** A multi-input circuit in which the output is 1 when any input is 1. Q=A+B
- **NOT** The output is 0 when the input is 1, and the output is 1 when the input is 0 .Q=$\bar{A}$
- **NAND** AND followed by INVERT.Q= $(A \cdot B)'$
- **NOR** OR followed by INVERT .Q= $(A+B)'$
- **XOR** the output of the Exclusive –OR gate, is 0 when its two inputs are the same and its output is 1 when its two inputs are different.Q= $A \oplus B = A'B + AB'$

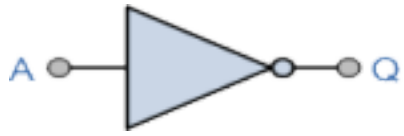
AND Operation:

Symbol	Truth table		
	A	B	Q=A.B
	0	0	0
	0	1	0
	1	0	0
	1	1	1

OR Operation:

Symbol	Truth table		
	A	B	Q=A+B
	0	0	0
	0	1	1
	1	0	1
	1	1	1

NOT Operation:

Symbol	Truth table	
	A	Q=Ā
	0	1
	1	0

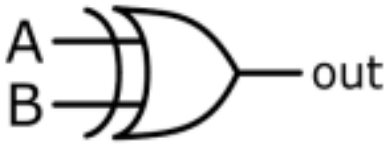
NOR Operation:

Symbol	Truth table		
	A	B	Q= (A+B)′
	0	0	1
	0	1	0
	1	0	0
	1	1	0

NAND Operation:

Symbol	Truth table		
	A	B	≡ Q= (A . B)′
	0	0	1
	0	1	1
	1	0	1
	1	1	0

XOR Operation:

Symbol	Truth table		
	A	B	Q= A⊕B = A′B+ AB′
	0	0	0
	0	1	1
	1	0	1
	1	1	0

Working Principles of the Circuit:

1. Logic States : The circuit operates using two voltages levels: High(5V) for logic 1 and Low(0V) for logic 0.
2. Signal Processing: Logic gates automatically process these binary input combinations to perform these binary combinations to perform logical addition(or), multiplication(and), inversion(not).
3. Verification: The circuit dynamically changes its output voltage to match the theoretical conditions predicted by the truth table.

5. Circuit Diagram:

6.Design and Implementation:

7.Experimental Procedure:

1. All ICs were placed on the breadboard across the center gap so that no two pins of an IC were shorted.
2. +5 V and GND from the trainer were connected to the breadboard power rails, and then to pin 14 and pin 7 of each IC.
3. Input switches A and B were wired to the input pins of the chosen gate in each IC.
4. The output pin of each gate was wired to a separate LED on the LED display (HIGH = LED on, LOW = LED off).
5. Power was turned on, and all input combinations 00, 01, 10 and 11 were applied one by one using the switches. The LED state was noted for each case.
6. The observed outputs were compared with the theoretical truth tables.

8.Truth Table and Observation

Operation	Input A	Input B	Theoretical Output	Observed Output
AND	13	12		11
OR	11	10		8
NOT	2			2
NOR	13	12		11
NAND	7	6		6
XOR	2	1		2

9. Experimental Results and Discussion

1. Did the experimental outputs match the theoretical outputs?
Ans: Yes.

2. Were there any unexpected observations?
Ans: No.

3. If any discrepancy occurred, what could have caused it?
Ans: If any discrepancy occurred, it could have been caused by loose connections, incorrect wiring, faulty components, or an unstable power supply.

4. What does the experiment demonstrate about the circuit's operation?
Ans:The experiment demonstrates that the logic gates operate according to their respective truth tables. The observed outputs matched the expected theoretical results for different input combinations.